Language-conditioned Spatial Affordance Refinement for Diffusion-based Grasp Detection

Bolin Tao
https://github.com/TBLboy/language-driven-grasp-detection

Abstract

Language-driven grasp detection requires grounding an instruction to the spatial region that affords a valid grasp, which semantic image-text alignment alone does not determine. We propose LSAR, a lightweight spatial affordance refinement module that injects a small-magnitude residual into the language-conditioned decoder feature before diffusion-based grasp generation. On a scene-disjoint 10,000-scene subset of Grasp-Anything++, LSAR increases the repeated-evaluation success rate from 23.5% (LGDM) to 33.9%, with 33.1% under a second LSAR training run, without changing the diffusion formulation or grasp representation.

1. Introduction

Language-driven grasp detection requires predicting a five-parameter grasp rectangle

$$G = (x, y, w, h, \theta) \tag{1}$$

for the object or part selected by a free-form instruction. Recognizing the referred object is not the same as locating the spatial region where a robot can apply a stable grasp: semantic alignment may identify the target, but it does not determine a feasible contact point. We start from LGDM, a diffusion-based model that conditions dense grasp maps on language. In the reference formulation, its ALBEF-conditioned spatial representation is not exposed to the decoder and the diffusion objective is not optimized. We define a clean LGDM configuration with both paths active, then introduce LSAR, a compact residual module applied before GG-CNN decoding. This adds spatial refinement without changing the pretrained encoders or the diffusion generator.

2. Related Work

Dense-map grasp detectors [1, 3] provide quality, angle, and width representations; LGD [4] extends this paradigm with Grasp-Anything++ [5] and diffusion-based conditional generation. Vision-language encoders such as ALBEF [2] align semantics, but semantic alignment does not encode where a stable grasp is physically possible. We therefore refine the language-conditioned spatial representation before grasp generation.

3. Method

3.1. Problem Setup

An input sample is (I, T), where I is an RGB image and T is a grasping instruction. Ground-truth rectangles are converted into dense maps pos, $\cos(2\theta)$, $\sin(2\theta)$, and normalized width; post-processing decodes the predicted dense maps into a rectangle.

3.2. Clean LGDM Baseline

LGDM reshapes an ALBEF-conditioned representation into $y_{\text{view}} \in \mathbb{R}^{8\times19\times19}$, providing a spatial language-conditioned feature compatible with the decoder bottleneck. In the reference configuration, this feature is not used and the diffusion loss is not optimized. Our clean baseline keeps the decoder structure but optimizes

$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{diffusion}} + \mathcal{L}_{\cos} + \mathcal{L}_{\sin} + \mathcal{L}_{\text{width}}. \tag{2}$$

Diffusion supervises the position map; the remaining terms align the decoded cosine, sine, and width maps.

3.3. LSAR

Semantic alignment does not identify which part of a target affords a grasp. LSAR combines the decoder feature F_v with y_{view} and learns a small-magnitude residual:

$$Z = \text{concat}\big(F_v,\ y_{\text{view}}\big), \tag{3}$$
$$H = \text{ReLU}\big(\text{conv}_{3\times3}(\text{GN}(\text{ReLU}(\text{conv}_{3\times3}(Z))))\big), \tag{4}$$
$$\Delta = s\,\text{conv}_{1\times1}(H), \tag{5}$$
$$F' = \text{ReLU}\big(F_v + \Delta\big). \tag{6}$$

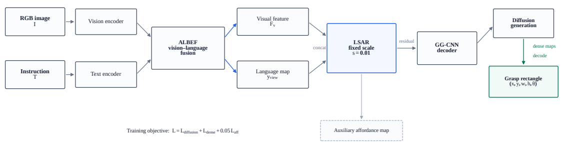

Figure 1. Overview of LSAR: ALBEF fusion, residual injection, GG-CNN decoding, and diffusion-based generation.

Table 1. Repeated 10-step evaluation on 2,000 validation stems. Mean and std are over three diffusion sampling runs of each checkpoint; † uses seed 43. Final LSAR uses scale 0.01.

Method	λ_{aff}	Success (%)
LGDM	—	23.5 ± 0.5
LSAR	0.0	32.7 ± 0.3
LSAR	0.10	30.3 ± 0.9
LSAR	0.05	$\mathbf{33.9} \pm 0.8$
LSAR†	0.05	33.1 ± 0.7

Here F_v and y_{view} provide visual and language cues; the learned correction Δ is rescaled by $s = 0.01$ so it remains small without overwhelming the visual feature. The module also emits a 19×19 affordance map $A = \text{conv}_{1\times1}(H)$ for the auxiliary term below.

3.4. Training Objective

The final objective is

$$\mathcal{L} = \mathcal{L}_{\text{base}} + \lambda_{\text{aff}}\,\text{MSE}\Big(A,\ \text{AdaptiveAvgPool}_{19}(\text{pos})\Big), \tag{7}$$

with $\lambda_{\text{aff}} = 0.05$ in the final method.

4. Experiments

4.1. Data and Protocol

10,000 unique-scene Grasp-Anything++ samples are split into 8,000 training / 2,000 validation with seed 42, so the splits do not share scenes. Training uses 15 epochs, batch 2, learning rate 10^{-3}, and weight decay 10^{-4}. Main evaluation uses 10-step respaced diffusion sampling. Correctness requires matching a ground-truth rectangle with IoU > 0.25 and angle error $< 30°$.

4.2. Results

Table 1 reports success rates. Unless noted, mean and std are computed over three diffusion sampling runs of the same checkpoint. LSAR improves LGDM from 23.5% to 33.9% under seed 42; a second LSAR seed 43 obtains 33.1%, suggesting the gain is not initialization-specific. The auxiliary weight is sensitive: $\lambda_{\text{aff}} = 0.05$ is best, removing it gives 32.7%, and $\lambda_{\text{aff}} = 0.10$ gives 30.3%; LSAR's principal gain is spatial conditioning. A 50-step verification preserves the model ordering. In the examples shown in Fig. 2, LSAR better localizes the highlighter and keychain parts; no qualitative claim beyond these examples is intended.

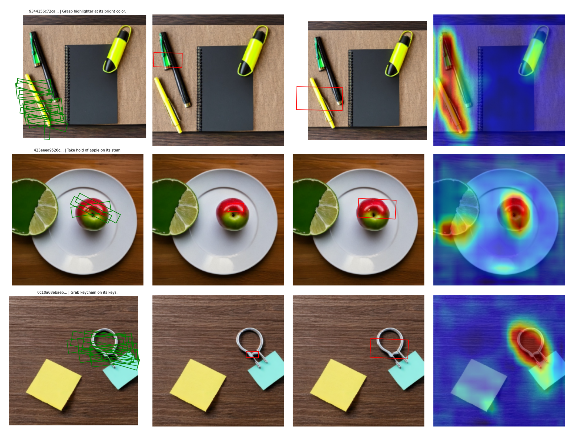

Figure 2. Paired qualitative results: input, ground truth, LGDM prediction, LSAR prediction, and LSAR affordance map.

5. Conclusion

Lightweight spatial refinement of language-conditioned features improves diffusion-based grasp prediction without changing the generative head (23.5% → 33.9%; second seed 33.1%). The study is limited to subset training and 10/50-step sampling; full-scale and real-robot evaluation remain future work.

References

[1] Sulabh Kumra, Shirin Joshi, and Ferat Sahin. Antipodal robotic grasping using generative residual convolutional neural network. In Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020.

[2] Junnan Li, Ramprasaath Selvaraju, Akhilesh Gotmare, Shafiq Joty, Caiming Xiong, and Steven Hoi. Align before fuse: Vision and language representation learning with momentum distillation. In Advances in Neural Information Processing Systems (NeurIPS), 2021.

[3] Douglas Morrison, Peter Corke, and Jürgen Leitner. Closing the loop for robotic grasping: A real-time, generative grasp synthesis approach. In Robotics: Science and Systems (RSS), 2018.

[4] An Dinh Vuong, Minh Nhat Vu, Baoru Huang, Nghia Nguyen, Hieu Le, Thieu Vo, and Anh Nguyen. Language-driven grasp detection. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.

[5] An Dinh Vuong, Minh Nhat Vu, Hieu Le, Baoru Huang, Binh Huynh, Thieu Vo, Andreas Kugi, and Anh Nguyen. Grasp-anything: Large-scale grasp dataset from foundation models. In Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2024.